Retrieval-Conditional Parsing Score (RCPS): Choosing Document Parsers by Retrieval, Not by Appearance

Anonymous ACL submission

Abstract

Practitioners selecting a document parser for retrieval-augmented generation (RAG) typically rank candidates by intrinsic quality metrics—boundary clarity, text-edit fidelity—on the assumption that cleaner parser output retrieves better. On Korean government documents the opposite holds, and the parser choice alone is decisive: in a six-parser, three-retriever evaluation, selecting by retrieval rather than by appearance swings Hit@1 by 2.8× (0.197 → 0.549), with the two cleanest-boundary parsers retrieving far below the messier vision–language ones. We make four contributions for production document-RAG. We characterise this parsing–retrieval disconnect in Korean and English—boundary clarity even anti-correlates with retrieval (Pearson $r = -0.81$, $n = 5$; sign-sensitive to the corpus)—and trace its more robust mechanism: under controlled semantic-noise perturbations, boundary clarity stays flat (MinerU, Qwen2.5-VL) or moves non-monotonically (GOT) while retrieval collapses in every family, indicating that intrinsic boundary metrics measure formatting rather than content (C1). A retriever-free coverage diagnostic attributes the gap to a single pipeline layer—20.2% of answers are absent from the parser output, a rate constant across eight chunkers—giving practitioners a rule computable before any retriever is run (C2). We propose RCPS, a retrieval-grounded selection protocol that requires no training and correctly ranks parsers and chunkers that intrinsic metrics misorder; an ablation shows it is not equivalent to single-embedder MRR (C3). Finally, we map the limits of parser-side training: a hidden-state auxiliary loss is sub-threshold and reference-free preference optimisation is negative, while best-of-K fidelity distillation yields a small but reproducible gain (OHR-Bench Hit@5 +1.22 pp), and a matched control shows the retrieval reward itself is unnecessary (C4). We release KoGovDoc-RAG, a Korean document-RAG benchmark, with a reference implementation of RCPS.

1 Introduction

Retrieval-augmented generation (RAG) is increasingly deployed over real-world document collections—government records, enterprise reports, technical manuals—that exist only as scanned or born-digital PDFs. Every such pipeline begins with a document parser that converts each page to text, which is then chunked, embedded, and retrieved. Because the parser sits at the head of the pipeline its errors cascade downstream, and production teams choose it with care. The practical question is *how.* Faced with many candidates—OCR engines, layout models, vision–language models—practitioners rank them by intrinsic parsing-quality metrics that reward clean-looking output: low text-edit distance and high boundary clarity (Zhao et al., 2025), on the assumption that cleaner output retrieves better.

On Korean government documents, that assumption fails, and which parser a team picks is decisive on its own. A practitioner ranking by intrinsic metrics would deploy MinerU (OpenDataLab, 2025), whose boundary-clarity score (0.72) is matched only by Marker; yet its retrieval Hit@1 is 0.20, against the 0.50–0.55 of three lower-scoring vision–language parsers—a 2.8× gap from parser selection alone. Boundary Clarity even anti-correlates with retrieval across these parsers (Pearson $r = -0.81$, $n = 5$; sensitive in sign to the corpus), but the robust signal is the mechanism: on a cross-domain English check (OHR-Bench, Zhang et al., 2025), injecting semantic noise collapses

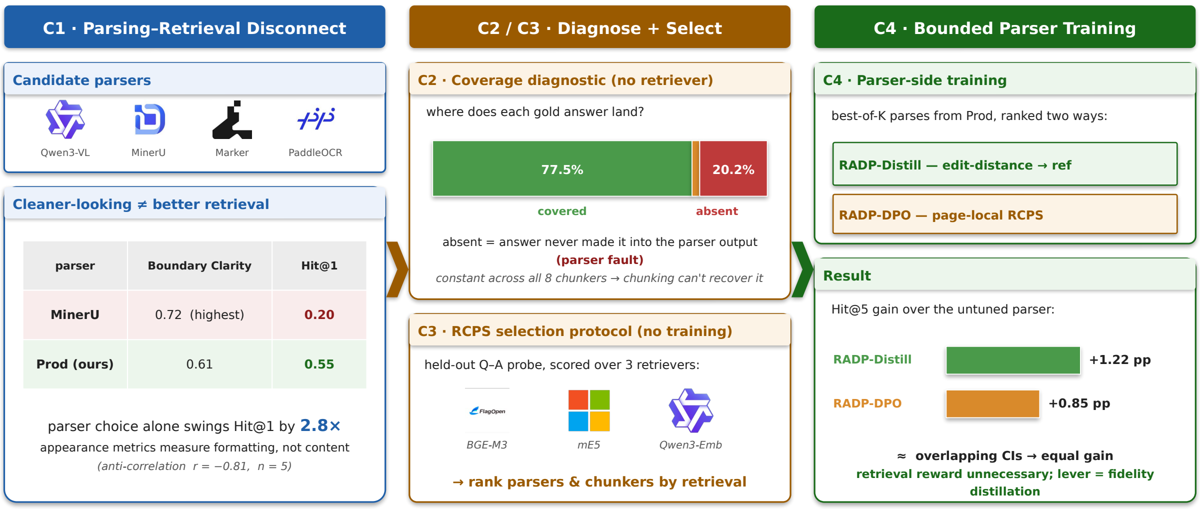

Figure 1: **Overview. (C1)** Selecting a document parser by appearance (Boundary Clarity) rather than retrieval is a trap: MinerU has the highest Boundary Clarity yet the lowest Hit@1, and parser choice alone swings Hit@1 by 2.8× ($r = -0.81$ at $n = 5$ is illustrative; the robust signal is the mechanism). **(C2)** A retriever-free coverage diagnostic localises the gap to the parser—20.2% of answers are absent from the parser output, constant across all eight chunkers. **(C3)** RCPS selects parsers and chunkers by retrieval over a held-out Q–A probe and three retrievers, with no training. **(C4)** Parser-side training is a bounded lever: best-of-K fidelity distillation (RADP-Distill) matches the retrieval-reward variant (RADP-DPO) (+1.22 vs +0.85 pp Hit@5, overlapping CIs), so the retrieval reward is unnecessary.

retrieval while Boundary Clarity barely moves, so the intrinsic metric cannot perceive the content corruption retrieval depends on. Selection by appearance optimises a quantity structurally blind to retrievability—at least for the scanned, layout-heavy documents we study.

We study how a team should instead select—and, where possible, improve—the parser, with an actionable procedure: select by RCPS, localise the bottleneck retriever-free, and honestly bound what parser-side training adds—in four contributions.

Figure 1 previews all four; the body develops each, including two points that surface only in the experiments—that RCPS is not equivalent to single-embedder MRR, and that two natural parser-training formulations, a hidden-state auxiliary loss and reference-free SimPO, are respectively sub-threshold and negative.

We release **KoGovDoc-RAG**, a Korean document-RAG benchmark of 663 question–answer pairs over 294 pages, together with a reference implementation of RCPS, the RADP-Distill checkpoint (the recommended parser-side lever), and the RADP-aux / RADP-DPO checkpoints used for the negative and controlled comparisons.

2 Related Work

Diagnosing the parsing–retrieval gap. Prior work documents that parser quality affects downstream retrieval but does not close the gap on the parser side. OHR-Bench (Zhang et al., 2025) evaluates the cascading impact of OCR noise on RAG; EnterpriseDocBench (Singh and Raj, 2026) reports a weak parsing-quality–retrieval correlation; and concurrent analyses (Sun et al., 2026) reach similar conclusions. Parsing benchmarks meanwhile rank parsers by intrinsic fidelity (Ouyang et al., 2025)—the signal C1 shows is misleading—and vision-based retrieval sidesteps text parsing entirely (Faysse et al., 2025), orthogonal to the text pipelines we target; Song (2026) likewise finds raw OCR degrades RAG. These diagnose the gap but give no reusable selection protocol or fault-localisation—what C2–C3 add.

Training-time methods, by pipeline layer. Existing retrieval-oriented training operates at every layer except the parser. Chunking methods set boundaries or filter chunks after parsing (Günther et al., 2024; Duarte et al., 2024; Zhao et al., 2024, 2025; Singh et al., 2025); embedding-side methods train the retriever contrastively (Conti et al., 2025; LMAR authors, 2025); and retrieval- or reader-side methods tune the retriever or generator (Nguyen et al., 2024; Chia et al., 2025; Yan et al., 2025). Parsers are themselves trained—recently with RL on edit-distance and layout rewards (Wang et al., 2025) or query-driven for efficient RAG (Wang et al., 2026)—but not on a retrieval signal; we ask what such a signal adds and find it bounded, the retrieval reward unnecessary over fidelity distillation (C4). Our primary contributions (C1–C3) sit upstream of any training.

3 The RCPS Protocol

3.1 RCPS: Retrieval-Conditional Parsing Score

RCPS ranks parsers by what downstream retrieval does with their output, not by how clean it looks. It is not a new similarity function but an evaluation protocol wrapping ordinary retrieval MRR in three choices: (i) it is *extrinsic*, scoring on the parsed corpus with a held-out Q–A probe rather than on text alone; (ii) it is *retriever-agnostic*, averaging over several embedders so the ranking does not hinge on the production one; and (iii) it uses *format-invariant relevance*, counting a chunk relevant iff its text contains the answer span, however formatted. The contribution is the protocol—what to measure, on what probe, how to judge relevance—not the scoring function, which is why RCPS is not single-embedder MRR (§4.3).

Given a parser P, a Q–A set $D = \{(q_i, a_i, \text{page}_i)\}$, a set of retrievers R, and cutoffs K, RCPS averages MRR across the cross-product:

$$\text{RCPS}(P) = \frac{1}{|R|\,|K|} \sum_{r \in R} \sum_{k \in K} \text{MRR@}k(r, P, D). \qquad (1)$$

We use $R = \{$BGE-M3 (Chen et al., 2024), multilingual-e5-large (Wang et al., 2024), Qwen3-Embedding-8B (Qwen Team, Alibaba, 2025a)$\}$ and $K = \{1, 5, 10\}$. A chunk is relevant iff its source page matches the answer's and the gold span is a substring under whitespace- and markdown-insensitive normalisation. In practice a team runs RCPS on a few hundred held-out Q–A, with no training, to choose a parser or chunker that intrinsic metrics rank incorrectly; Figure 4 sketches the protocol and the failure each choice removes. The implementation is released.

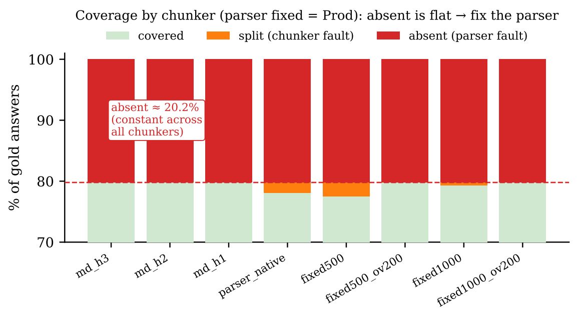

Figure 2: The coverage diagnostic (C2): with the parser fixed (Prod) and the chunker varied, each gold answer is *covered*, *split* (a chunker fault, recoverable with overlap), or *absent* (a parser fault, unrecoverable by chunking). The absent rate is ≈ 20.2% and essentially constant across all eight chunkers, while split is at most 2.3% (§4.2)—chunking cannot close a gap the parser already lost, so the fix must be parser-side.

3.2 Coverage Diagnostic — Locating the Gap (Parser vs Chunker)

RCPS scores the parser, chunker, and retriever jointly, so a low score does not localise the fault. We separate the parser from the chunker with a retriever-free diagnostic: holding the parser output fixed and varying the chunker, we classify each gold answer as *covered* (inside a single chunk, retrievable); *split* (present in the page text but cut across chunks—a chunker fault, recoverable with overlap); or *absent* (not in the parser output at all—a parser fault, unrecoverable by chunking). Coverage is the ceiling on retrieval, since split and absent answers score zero; the split/absent breakdown attributes the gap to a pipeline layer and decides whether a parser-side fix can help (§4.2). Relevance reuses RCPS's format-invariant matching.

4 Experiments

4.1 Setup

We construct **KoGovDoc-RAG**: 663 Q–A pairs over 294 pages of Korean government documents, generated with the OpenAI `gpt-5.4-2026-03-05` model and verified by an LLM-as-judge against the teacher-distilled reference markdown (Qwen3-VL-30B pseudolabels; 100 eval Q–A sampled, 94/100 accept). For RADP-aux's full-scale training we additionally generate 6,164 Q–A on the 2,667-page Prod training set. Cross-domain replication uses OHR-Bench (Zhang et al., 2025) (seven domains, 2,264 verbatim-answerable Q–A; its Law + Manual subset of 1,043 Q–A is used only for the data-mix-sensitivity comparison in §4.2) across the three released parser outputs (gt, MinerU, Qwen2.5-VL) plus twelve controlled noise perturbations.

Throughout, **Prod** denotes our production parser, a Qwen3-VL-2B model (Qwen Team, Alibaba, 2025b) fine-tuned for Korean document parsing; all training deltas are measured against it. RADP-aux uses a 73-page held-out fold (202 Q–A); RADP-DPO, SimPO, and the §4.4 analysis use the combined 242-page fold (train ∪ eval, 663 Q–A), with preference pairs built only from the 169-page train fold. The same 663 Q–A occupy 242 of the 294 KoGov pages; the coverage diagnostic (§4.2) runs over Prod's full 294-page output, the extra 52 Q–A-free pages acting as distractors. Parses for the 12-variant comparison (§4.4–§4.4) use deterministic decoding (temperature 0, max 1536 tokens). We fine-tune with LoRA ($r = 8$, $\alpha = 32$), sweeping $\lambda \in \{0, 0.1, 0.3, 0.5\}$ for RADP-aux ($\lambda = 0$ drops the contrastive term; the aux recipe itself shifts output from Prod, Table 5, so the sweep is read within the aux family). RCPS uses the three retrievers and cutoffs of §3.1; we report paired percentile-bootstrap 95% CIs (1,000 resamples), preserving Q–A pairing across systems.

4.2 The Parsing–Retrieval Disconnect and Coverage Diagnostic (C1–C2)

Korean government documents (Figure 3; full grid in Table 6, Appendix D). RCPS spans 0.07–0.58 across the six parsers: the vision–language parsers cluster at the top (0.53–0.58) and the OCR systems trail (0.07–0.21). Within that top cluster the 30B teacher ranks first (RCPS 0.584), with the deployable 2B Prod a negligible 0.001 behind (0.583); Figure 3b therefore contrasts Prod, the model a team would ship, against the appearance-ranked MinerU. The intrinsic Boundary Clarity metric (Zhao et al., 2025) anti-correlates with RCPS at Pearson $r = -0.81$ ($n = 5$, excluding PaddleOCR, whose Boundary Clarity is undefined because MoC detects zero boundaries in its output). The two cleanest-boundary parsers (MinerU, Marker; BC $\approx$ 0.72) land near the bottom, while the lower-BC vision–language parsers retrieve best. Marker is evaluated on a 38-page subset; dropping it leaves the anti-correlation unchanged in sign and magnitude ($n = 4$, $r = -0.74$). MoC's companion metric, Chunk Stickiness, is similarly disconnected ($r = +0.26$, $n = 5$; with CS oriented so that lower is more cohesive, this is the same direction as BC—more cohesive intrinsic structure tracks worse retrieval). Neither intrinsic axis predicts RCPS.

A single-domain anti-correlation could be a quirk of one language or document type, so we test cross-domain.

Cross-domain: the mechanism. Across all seven OHR-Bench domains (2,264 Q–A) we evaluate 15 parser-output variants: three released parser outputs (gt, MinerU, Qwen2.5-VL), three formatting-noise perturbations, and nine semantic-noise perturbations (GOT, MinerU, Qwen2.5-VL $\times$ mild, moderate, severe). Within each semantic-noise family, Boundary Clarity fails to track noise severity while RCPS collapses (per-variant values in Table 7, Appendix E). For the MinerU family, BC stays in 0.71–0.74 from clean to severe while RCPS falls from 0.50 to 0.24 (-51%); GOT shows the same pattern (RCPS 0.38 $\rightarrow$ 0.26) and Qwen2.5-VL is more noise-robust (0.47 $\rightarrow$ 0.43, -8%). Semantic noise that destroys retrievable content does not lower Boundary Clarity: the intrinsic metric sees formatting, not content.

The aggregate cross-variant correlation is sensitive to the document mix: Pearson BC$\leftrightarrow$RCPS across all 15 variants is -0.35 on Law + Manual alone but $+0.25$ on the full seven-domain corpus. We therefore report the per-family mechanism, which reproduces in every domain, as the robust finding, and treat the cross-variant scalar—which conflates parser families of differing noise-robustness—as illustrative only.

Locating the gap: parser or chunker? (Figure 2, Table 1). The disconnect shows that intrinsic metrics mislead, but not which pipeline layer is at fault. Applying the §3.2 classification with no retriever, on Prod's out-

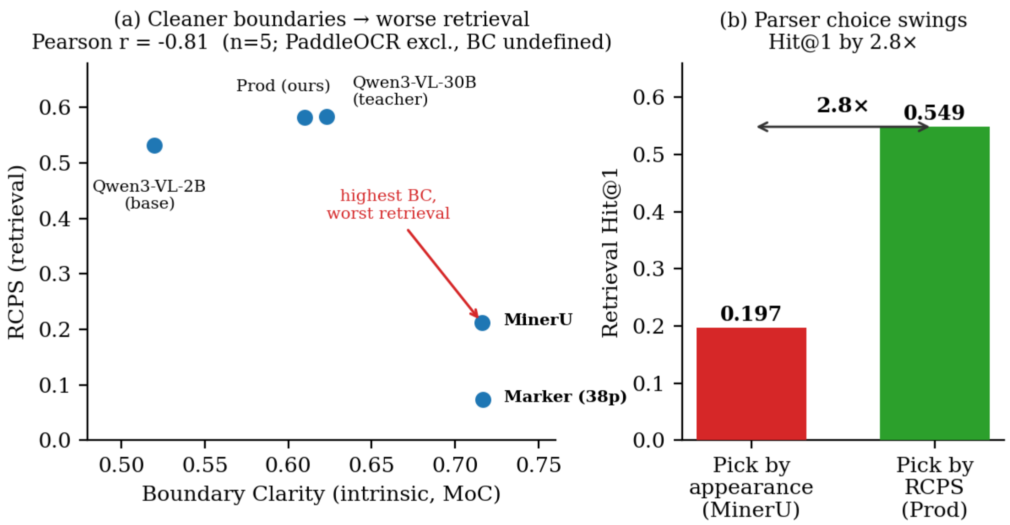

Figure 3: The parsing–retrieval disconnect on Korean government documents. (a) Boundary Clarity (intrinsic, MoC) anti-correlates with RCPS across parsers (Pearson $r = -0.81$, $n = 5$; PaddleOCR excluded as its BC is undefined): the cleanest-boundary parsers MinerU and Marker land near the bottom, while the lower-BC vision–language parsers retrieve best. (b) Choosing the parser by RCPS (Prod) rather than by appearance (MinerU) changes retrieval Hit@1 by 2.8× (0.197 → 0.549).

Chunker	covered	split	absent
md_h3	79.8%	0.0%	20.2%
parser_native	78.1%	1.7%	20.2%
fixed500	77.5%	2.3%	20.2%
fixed500_ov200	79.8%	0.0%	20.2%

Table 1: Answer-coverage diagnostic on Prod's output (294 pages, 663 KoGov Q–A; text matching only, no retriever). The absent rate (a parser fault) is constant at 20.2% across all eight chunkers (four representative rows shown), the required boundary-independence check; the split rate (a chunker fault) is at most 2.3% and vanishes under overlap. Chunkers: md_h{1,2,3} = markdown-header splitting at heading depth k; parser_native = the parser's own segmentation; fixed N[_ov$M$] = fixed N-character windows, optionally with M-character overlap.

put (294 pages, 663 Q–A) 20.2% of answers are absent and at most 2.3% are split, and the absent rate is constant across all eight chunkers we test—exactly the boundary-independence a parser fault must show. The gap is therefore a parser problem: roughly one answer in five is never produced, so no re-chunking can recover it, and a parser-side intervention is the correct lever. This both motivates the training in §4.4 and yields the practitioner rule: run this diagnostic first, and if absent dominates, fix the parser; if split dominates, fix the chunker.

4.3 RCPS Selects Both Parsers and Chunkers (C3)

A selection protocol must separate the alternatives a practitioner actually compares, across both knobs they control. For **parsers**, the six-parser grid of §4.2 (Figure 3, Appendix D) is itself an RCPS ranking: it tells a team that Prod (RCPS 0.583, Hit@1 0.55) beats MinerU (0.21, 0.20) by 2.8×, the ordering Boundary Clarity inverts. For **chunkers**, on a fixed Prod output RCPS separates four strategies cleanly—markdown-header (md-h3, RCPS 0.593) > parser-native (0.583) > LumberChunker (Duarte et al., 2024) (0.557) > fixed-size (0.535)—whereas intrinsic boundary metrics rank them inconsistently, or rank fixed-size highest because its boundaries are clean by construction. Because RCPS is retriever-averaged and format-invariant, one probe set of a few hundred Q–A ranks both the parser choice and the chunker choice a team makes independently.

RCPS is not single-embedder MRR (Table 2). Stripping the three protocol choices from the six-parser KoGov grid shows they are not cosmetic. Dropping retriever-averaging—scoring on a single embedder (BGE-M3)—inverts the top parser choice: naive single-embedder MRR ranks Prod first, while full

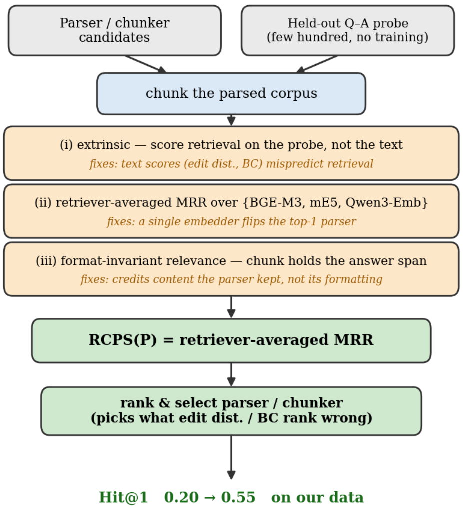

Figure 4: RCPS as a protocol, not a new scoring function (C3). It wraps ordinary retrieval MRR in three choices, each removing a failure mode of intrinsic metrics: (i) a held-out Q–A probe rather than the text, since text scores (edit distance, BC) mispredict retrieval; (ii) retriever-averaging, since a single embedder can flip the top-ranked parser (Table 2); and (iii) format-invariant relevance, crediting the content a parser keeps. Run with no training, it ranks the parser and chunker a team would deploy.

Protocol configuration	Top pair	τ vs RCPS
naive MRR (1 embedder, fmt-sensitive)	Prod $\succ$ 30B	0.87
+ retriever-averaging	30B $\succ$ Prod	1.00
+ format-invariant only	Prod $\succ$ 30B	0.87
full RCPS (3 retrievers, fmt-invariant)	30B $\succ$ Prod	—

Table 2: RCPS protocol ablation on the KoGov six-parser grid (only the inverted top pair is shown; the lower ranks Qwen3-VL-2B $\succ$ MinerU $\succ$ PaddleOCR $\succ$ Marker are identical across all configurations). Retriever-averaging is the single choice that flips the top pair (rows 2,4 vs 1,3); format-invariance shifts scores but not order. Kendall τ is over the full six-parser ranking; "30B" is the Qwen3-VL-30B teacher.

RCPS ranks the Qwen3-VL-30B teacher first (Table 2). Format-invariant matching, by contrast, shifts every parser's score by about 0.02–0.03 but does not reorder the grid. The inversion is between two near-tied parsers (RCPS 0.584 versus 0.583), so the operational claim is precise: single-embedder MRR cannot reliably resolve the top parser a team would deploy—exactly the decision RCPS is run to make. The ordering disagreement (Kendall $\tau = 0.87$) substantiates the claim that RCPS is a protocol, not a relabelled MRR.

4.4 Parser-Side Training Is a Bounded, Reward-Agnostic Lever (C4)

When the coverage diagnostic points to the parser (§4.2), we train it on a retrieval signal—a family we call Retrieval-Aware Document Parsing (RADP)—in two natural ways. **(a) Hidden-state auxiliary loss (RADP-aux):** an InfoNCE term aligns the parser's pooled answer-span hidden state with the frozen BGE-M3 embedding of the gold chunk ($\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{parse}} + \lambda\,\mathcal{L}_{\text{contrast}}$); this differentiable surrogate reaches the deployed markdown only through $\mathcal{L}_{\text{parse}}$. **(b) Discrete-output DPO (RADP-DPO):** we optimise the markdown directly (Rafailov et al., 2023)—sampling K parses per page, scoring each by a page-local RCPS, forming preference pairs (gap ≥ 5 pp), and training with a LoRA-toggle reference (Hu et al., 2022) (π_θ adapter-on, π_{ref} adapter-off), the reward sharpened over three milestones R1→R2→R3 (Appendix A); reference-free SimPO (Meng et al., 2024) is a control. **(c) Reward-agnostic control (RADP-Distill):** the identical best-of-K pipeline but ranking candidates by edit-distance to the reference markdown instead of RCPS.

Of the three directions, only discrete-output preference training produces a deployable effect—the hidden-state auxiliary loss is sub-threshold and reference-free SimPO is uniformly negative (Appendix B). On KoGov, RADP-DPO improves Hit@5 by +1.96 to +2.11 pp, but at $n = 663$ every two-sided interval spans zero, so we treat KoGov as exploratory and rely on the pre-specified OHR-Bench replication, where training signal, metric, and language are disjoint. There the representative variant R2 gives Hit@5 +0.85 pp $[+0.35, +1.43]$, two-sided significant across all seven domains (Table 4).

The retrieval reward is unnecessary. A matched RADP-Distill control—identical but ranking pairs by edit-distance to the reference markdown rather than RCPS—reproduces the gain (OHR Hit@5 +1.22 pp $[+0.35, +2.15]$; KoGov +2.61 pp), indistinguishable from

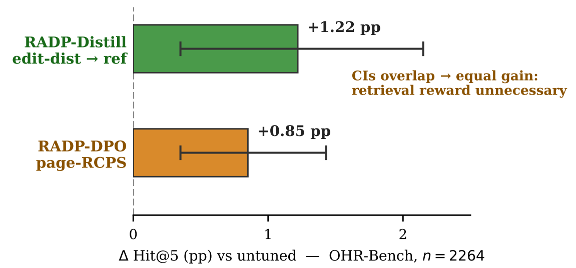

Figure 5: Parser-side training (C4): OHR-Bench Hit@5 gain over the untuned parser for the two ways of ranking best-of-K candidates. Ranking by edit-distance to the reference markdown (RADP-Distill, +1.22 pp) and by page-local RCPS (RADP-DPO, +0.85 pp) are both significant and their confidence intervals overlap—a statistically indistinguishable gain. The retrieval reward is therefore unnecessary; the deployable lever is fidelity distillation (3-retriever avg, n=2264).

RADP-DPO at the powered endpoint (Figure 5). The lever is therefore fidelity distillation, not the retrieval reward—a cheaper route to the same target.

The mechanism is text fidelity, not chunk boundaries. Across the 12 systems (Appendix C, Table 5), every DPO variant cuts TextNED (normalised edit distance to the reference markdown) from Prod's 0.240 to 0.16–0.19 and RADP-Distill cuts it furthest (0.158), while the MoC chunking signature (BC 0.60–0.66, CS $\approx$ 0.47) does not move; RADP-aux, which instead *raises* TextNED, stays sub-threshold. The gain comes from *what* is parsed, not *how* it is chunked—§4.2's intrinsic-metric blindness seen from the training side—and replicates cross-domain (English R2 TextNED −1.36%, significant). RADP-DPO is thus best read as best-of-K self-distillation: training amortises into the weights the higher-fidelity parses the model could already occasionally produce.

5 Discussion and Conclusion

Selection, not training, is the headline. The largest and most certain effect here is not a training method: choosing a parser by retrieval (RCPS) rather than by intrinsic metrics changes Hit@1 by 2.8× (§4.2). A team that adopts only the RCPS protocol already avoids shipping the parser that looks best yet retrieves near the bottom. Parser-side training is a secondary, bounded lever of about 1 pp Hit@5, and the retrieval reward buys nothing over plain fidelity distillation (§4.4).

Why the diagnosis and the mechanism agree. C1 finds the cleanest-boundary parser retrieves among the worst; §4.4 finds training helps by producing text closer to ground truth. Both follow from splitting "clean" into two axes—Boundary Clarity measures formatting, TextNED measures content fidelity (whether the text holds the answer span). MinerU is formatting-clean but loses content; training improves fidelity while leaving the formatting signature unchanged, which is exactly why intrinsic boundary metrics mispredict retrieval. This also refines our starting hypothesis: we expected training to move chunk boundaries, but what moves is text fidelity.

Playbook and conclusion. For practitioners: (1) evaluate parsers with RCPS, not intrinsic metrics—a few hundred held-out Q–A, no training, reorder parsers as the retriever sees them (on our data a 0.20 → 0.55 Hit@1 decision); (2) if you train, distil the discrete output toward clean ground-truth text—the auxiliary loss and reference-free SimPO are sub-threshold, and a retrieval reward is unnecessary; (3) the gain concentrates on factoid queries and is neutral on tabular ones, so layout-heavy query mixes should instead look to the chunker or embedder. We documented the parsing–retrieval disconnect in two languages, proposed RCPS for selecting parsers and chunkers by retrieval rather than appearance, and bounded what parser-side training adds. Code, data, and checkpoints are released.

Limitations

Single primary language and statistical power. The C1 diagnostic is strongest in Korean (n = 5, r = −0.81). On English OHR-Bench the aggregate cross-variant scalar is sign-sensitive to document mix (r = −0.35 on Law + Manual, +0.25 on the full seven-domain corpus), so directional consistency holds only for the per-family noise mechanism (§4.2), which replicates in every domain—not for the scalar correlation. The English evidence is also built on three real parser outputs plus twelve controlled perturbations rather than fifteen independent parsers. With these sample sizes the correlations are illustrative rather than inferential.

Q–A generation and construct validity. The KoGov reference markdown is itself pseudo-ground-truth distilled from a Qwen3-VL-30B teacher (with manual de-noising of contaminated samples), and the Q–A were LLM-generated and checked by an LLM-as-judge against that reference (94/100 accept on a 100-Q–A stratified sample); neither the reference parses nor the queries were fully human-verified. Synthetic queries and pseudo-reference bound what RCPS can claim absolutely, but the exposure is narrow: our comparative findings—the 2.8× parser-choice swing and every parser/chunker ranking—hold the Q–A set fixed across systems and so are internally valid regardless of how queries were generated, and the confirmatory cross-domain test (§4.4) uses OHR-Bench's own externally curated Q–A. We release the frozen KoGov eval set for audit.

KoGov is exploratory; OHR-Bench is confirmatory. The headline KoGov cell (R2 +1.96 pp, Table 3) has a paired 95% interval of $[-1.06, +5.03]$—two-sided non-significant and selected from a multi-cell scan—so we claim no two-sided significance on KoGov. The confirmatory evidence is the pre-specified OHR-Bench replication (R2 +0.85 pp $[+0.35, +1.43]$, $n = 2{,}264$). A larger KoGov fold ($\geq$ 1,500 Q–A) would be needed for a powered two-sided result.

Candidate pool and untested paradigms. Preference pairs are sampled from Prod itself; an alternative pool (e.g. chosen/rejected pairs from other parsers) might shift the mechanism away from GT-fidelity, which we did not test. We also test only the hidden-state auxiliary loss, discrete-output DPO with a reference policy, and reference-free SimPO; token-level RL, multi-task interleaving, curriculum schedules, and chunk-granularity oracle distillation are untested. Finally, RADP-DPO is a parser-layer intervention; chunker-, embedder-, and reranker-side training on RCPS-graded chunks are complementary directions left to future work.

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A RADP-DPO Milestone Construction

For each train page we sample K alternative parses from the production parser, chunk and score each by a page-local RCPS (the page's questions against the parse's own chunks plus distractors), and form preference pairs (chosen = higher-RCPS parse, gap ≥ 5 pp). The reward is sharpened across three milestones: **R1** ($K = 2$ parses, uniform distractors, $\beta = 0.1$) $\rightarrow$ **R2** (a warm-started iterative round, $\beta = 0.05$) $\rightarrow$ **R3** ($K = 14$ parses with a full-corpus hard-negative pool—the other-page chunks closest to the page's queries—widening the candidate-score gap from about 0.05 to 0.56). We train with a LoRA-toggle reference (π_θ = LoRA on, π_{ref} = LoRA off):

$$\mathcal{L}_{\text{DPO}} = -\log\sigma\Big(\beta\big[(\log\pi_\theta(c) - \log\pi_\theta(r)) - (\log\pi_{\text{ref}}(c) - \log\pi_{\text{ref}} \tag{2}$$

The reference-free SimPO control ($\beta = 2.0$, $\gamma = 1.0$) removes the reference policy entirely.

B KoGov DPO Progression, RADP-aux Sweep, SimPO, and Robustness

On the 242-page KoGov fold (10,000-resample paired bootstrap), the three RADP-DPO milestones improve Hit@5 on parser-native chunking over Prod along the reward-sharpening axis: R1 +2.06 pp [−0.96, +5.13], R2 +1.96 pp [−1.06, +5.03], R3 +2.11 pp [−0.96, +5.13] (all $P[\Delta{>}0] \approx 0.90$). At $n = 663$ every two-sided interval spans zero, so these are exploratory: the Hit@5 / parser-native cell was selected from a multi-cell scan over (chunker × retriever × k) with no family-wise correction. The reward-agnostic RADP-Distill control, evaluated on the same fold, reaches +2.61 pp and is the headline; its powered confirmation is the OHR-Bench result (Table 4, +1.22 pp).

RADP-aux is sub-threshold. The auxiliary loss peaks at $\lambda = 0.1$ (+1–2 pp RCPS) then declines monotonically as the contrastive objective competes with $\mathcal{L}_{\text{parse}}$ over the same

Variant (vs Prod = 0.6863)	Hit@5	ΔHit@5 [95% CI]	$P[\Delta{>}0]$	ΔRCPS
RADP-DPO-R3	0.7074	+2.11 [−0.96, +5.13]	0.913	+1.72
RADP-DPO-R1	0.7069	+2.06 [−0.96, +5.13]	0.907	+0.57
RADP-DPO-R2	0.7059	+1.96 [−1.06, +5.03]	0.897	+0.47
RADP-SimPO (ctrl)	0.6793	−0.70 [−3.77, +2.31]	0.321	−1.56

Table 3: RADP-DPO milestone progression and the SimPO control on parser-native chunking, 242-page / 663-Q–A fold, 10k paired percentile bootstrap, all against the same Prod baseline (Hit@5 = 0.6863). Macro Hit@k averages the three retrievers of §3.1. All three milestones are strong directional positives ($P[\Delta{>}0] \geq 0.90$) but two-sided non-significant at this fold size; RADP-SimPO is uniformly negative. The RADP-Distill headline (KoGov ΔHit@5 +2.61 pp) is in §4.4 and Table 4. Across three Prod seeds the standard deviation of the mean ΔHit@5 is 0.90 pp.

Δ vs Prod (pp)	Hit@1	Hit@5	Hit@10	MRR@10	nDCG@5
RADP-Distill	**+0.88**	**+1.22**	**+1.32**	**+1.01**	**+1.05**
RADP-DPO-R2	+0.53	+0.85	+0.81	+0.70	+0.74
RADP-DPO-R3	+1.31	+1.03	+0.81	+1.17	+1.15

Table 4: OHR-Bench cross-domain Δ vs Prod, in pp (2,264 Q–A over seven English domains; three-retriever macro; 1,000-resample paired bootstrap). The Hit@5 deltas are two-sided significant (95% CIs: RADP-Distill [+0.35, +2.15], R2 [+0.35, +1.43], R3 [+0.24, +1.84]). The other columns are monotone re-cuts of the same per-system ranked list, reported only because practitioners use different cutoffs. RADP-Distill leads at the primary metric Hit@5; R3 reaches higher Hit@1 and MRR@10 point estimates but with overlapping CIs, so the retrieval reward is unnecessary at the powered endpoint (§4.4).

LoRA parameters; every Δ-versus-control interval includes zero and the one-sided $P[\Delta{>}0]$ never approaches 0.90. The hidden-state route does not reach the deployed markdown—only the discrete output does.

SimPO is negative. The reference-free length-normalised variant gives uniformly negative deltas (−0.7 to −1.7 pp Hit@5), indicating the DPO reference policy is load-bearing: without it the parser drifts from the production distribution before any preference signal can compound.

Robustness. The headline RADP-Distill OHR-Bench gain survives two checks. Because it ranks candidates by edit-distance and uses no retriever, it cannot overfit any embedder; its Hit@5 gain is near-uniform across the three (BGE-M3 +1.06, ml-e5-large +1.10, Qwen3-Emb +1.50 pp). And it concentrates on text-evidence (factoid) queries (+1.19 pp) while near-neutral on table-evidence ones (+0.32 pp), consistent with the text-fidelity mechanism.

C Chunk-Level Mechanism Statistics

We measure four chunk-level statistics on the 242-page fold for all systems in Table 5: MoC Boundary Clarity (BC, adjacent-chunk discontinuity), MoC Chunk Stickiness (CS, within-chunk cohesion), normalised edit distance to the reference markdown (TextNED, character-level Levenshtein distance ÷ longer-string length), and chunking shape.

The signal. All three RADP-DPO milestones reduce TextNED from Prod's 0.240 to 0.163–0.185, a 23–32% move toward the reference markdown; R1 and R2 achieve the largest reductions (0.167, 0.163). RADP-aux instead increases TextNED (0.318–0.626) because its hidden-state objective degrades surface text. TextNED is positively associated with Hit@5 (R3 the one DPO-cluster exception), and RADP-Distill reduces it furthest (0.158) while leading on retrieval—the cleanest demonstration that text fidelity, not the retrieval reward, is the lever.

Cross-domain replication. On all 4,040 English OHR-Bench pages, zero-shot Prod reaches TextNED 0.192—below its 0.240 on Korean, so English leaves less fidelity headroom. R2 reduces it to 0.1894 (a −0.0026 delta, −1.36%, 95% CI [−0.0043, −0.0010], two-sided significant). R3 pushes the English value marginally lower (0.184) yet does not improve KoGov fidelity, which is why R2 is the

Variant	len	c/pg	clen	BC↑	CS↓	TextNED↓
Prod (ref)	1385	4.82	274	0.630	0.474	0.240
RADP-Distill	1597	5.29	291	0.641	—	**0.158**
aux λ=0.0	799	2.96	245	0.553	0.473	0.626
aux λ=0.1	1380	4.05	321	0.652	0.484	0.423
aux λ=0.3	1930	6.73	272	0.470	0.475	0.332
aux λ=0.5	2035	5.82	327	0.556	0.470	0.318
DPO-R1	1606	4.97	311	0.646	0.474	**0.167**
DPO-R2	1610	4.83	321	0.647	0.476	**0.163**
DPO-R3	1514	4.88	300	0.656	0.485	0.185
SimPO	1703	5.31	305	0.601	0.470	0.186

Table 5: Chunk-level mechanism statistics on the 242-page fold (len = parse length, c/pg = chunks per page, clen = chunk length). RADP-Distill attains the lowest TextNED of all (0.158 < R2's 0.163 < Prod's 0.240) with BC unchanged (0.641, within Prod's range), confirming that text fidelity, not chunk structure, is the operative axis (CS was not measured for RADP-Distill, which uses a fixed chunking scheme). Among retrieval-reward variants, R2 has the lowest TextNED (0.163); R3's KoGov TextNED (0.185) does not beat it.

headline variant.

The non-signal. The MoC chunking signature is unchanged: BC for every DPO variant lands in 0.60–0.66 (Prod 0.63) and CS near 0.47–0.49 (Prod 0.474); chunks-per-page (∼5) and chunk length (300–321) stay close to Prod's. The gain comes from what is parsed, not how chunks are split.

Secondary patterns. The gain concentrates on text-evidence (factoid) queries (+1.19 pp)—where the verbatim answer span must land inside a chunk—and is near-neutral on table-evidence ones (+0.32 pp). Because RADP-Distill ranks by edit-distance and uses no retriever, its gain is near-uniform across the three embedders, confirming a parser-level effect rather than an embedder artefact. RADP-aux stays sub-threshold because its hidden-state signal reaches the deployed markdown only through diffuse $\mathcal{L}_{\text{parse}}$ back-flow, never localising to the surface tokens the retriever rewards.

D KoGov Parser Grid

E OHR-Bench Noise-Family Grid

Parser	BC	CS	RCPS	Hit@1
Qwen3-VL-30B (teacher)	0.623	3.38	**0.584**	0.545
Prod (ours, 2B)	0.610	3.07	0.583	0.549
Qwen3-VL-2B (base)	0.520	3.74	0.532	0.500
MinerU	0.716	2.81	0.212	0.197
PaddleOCR	—	3.46	0.140	0.125
Marker (38p)	**0.717**	3.41	0.073	0.068

Table 6: The §4.2 disconnect grid, visualised in Figure 3: KoGov, BC versus RCPS, Pearson $r = -0.81$ ($n = 5$, excluding PaddleOCR, whose Boundary Clarity is undefined). Chunk Stickiness (CS, MoC; lower = more cohesive) is similarly uninformative ($r = +0.26$, $n = 5$); its scale differs from the per-variant CS in Table 5. RCPS is averaged over three retrievers.

Family	BC (clean→severe)	RCPS (clean→severe)
MinerU	0.735/0.708/0.715/0.712	0.496/0.413/0.351/0.243
Qwen2.5-VL	0.619/0.614/0.616/0.610	0.467/0.462/0.460/0.429
GOT	–/0.634/0.495/0.650	–/0.383/0.338/0.257

Table 7: Per-family Boundary Clarity and RCPS across semantic-noise severity (clean/mild/moderate/severe) on the seven OHR-Bench domains—the source for the §4.2 noise numbers. Within each family BC stays roughly flat while RCPS falls (MinerU −51%). GOT has no clean baseline; the GT output and three formatting-noise variants enter only the aggregate scalar (§4.2) and are in the released grid.